

# Market Structure and Rapeseed Volatility

March 2026

**Data.** 2,094 daily price bars (Jan 2018–Mar 2026) merged with 407 weekly Euronext positioning reports covering four participant categories (investment firms, funds, other financials, commercials). Positioning dates are as-of Friday; reports are published the following Wednesday. Data is forward-filled to the next available positioning date to avoid leakage.

## 1 The Question

Does the *distribution* of open interest across participant categories predict higher short-term volatility—particularly around roll and pre-expiry windows?

## 2 Market Structure Predicts Volatility

Two variables computed from weekly Euronext positioning reports carry forward-looking volatility information.

**HHI (Herfindahl–Hirschman Index):** the sum of squared gross-share proportions across the four participant categories:

$$\text{HHI} = s_1^2 + s_2^2 + s_3^2 + s_4^2$$

where  $s_i$  is category  $i$ 's share of total gross open interest. In this dataset HHI ranges from 0.35 to 0.77. Its 13-week z-score (HHIz13) captures whether concentration is rising or falling relative to recent history.

**Herdling:** for each pair of participant categories  $(i, j)$ , we compute the rolling Pearson correlation of their weekly net-position changes  $\Delta N_i, \Delta N_j$  over the preceding 8 weekly observations. The herding index is the average across all  $\binom{4}{2} = 6$  pairs:

$$\text{Herdling}_t = \frac{1}{6} \sum_{i < j} \rho_{8w}(\Delta N_i, \Delta N_j)$$

When herding is high, participants move together; when it is low (or negative), they disagree. Herding is only weakly correlated with HHI ( $r = 0.25$ ): HHI measures the static imbalance in how open interest is distributed, while herding measures the dynamic agreement in how participants change their positions. Both carry independent information about the volatility outlook.

**Garman–Klass (GK) volatility** is used throughout as the primary daily vol measure. All OLS regressions include a turnover ratio (volume / open interest) as a liquidity control.

### 2.1 Economic Magnitude (Quintile Analysis)

Sorting daily observations by herding quintile reveals a monotonic, economically large relationship:

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<sup>1</sup>Quintiles are formed globally over the full sample and therefore use future information for bin edges. This is standard for exploratory analysis but means the quintile magnitudes are descriptive, not predictive.

**Table 1:** Volatility outcomes by herding quintile<sup>1</sup>

| Herding quintile     | 20d fwd rvol (%) | GK-vol (%) | Daily range (%) |
|----------------------|------------------|------------|-----------------|
| Q1 (most divergent)  | 1.54             | 1.23       | 2.02            |
| Q2                   | 1.38             | 1.16       | 1.93            |
| Q3                   | 1.33             | 1.11       | 1.83            |
| Q4                   | 1.32             | 1.05       | 1.69            |
| Q5 (most correlated) | 1.22             | 0.93       | 1.54            |
| Q1/Q5 ratio          | 1.26             | 1.32       | 1.31            |

When participants diverge (Q1), 20-day forward realised vol is 26% higher and GK-vol is 32% higher than when they herd (Q5). For HHIz13, top-quintile (rising concentration) days show 48% higher 20-day forward vol and 22% higher GK-vol than the lowest quintile.

## 2.2 Event Study: Volatility Around Contract Expiry

For each of 31 contract expiries we extract a  $[-20, +5]$  trading-day window, split by regime (above/below median) at expiry.

**Table 2:** GK volatility around expiry by HHIz13 regime

| Days to expiry | High HHIz13 (%) | Low HHIz13 (%) |
|----------------|-----------------|----------------|
| -20            | 1.36            | 1.17           |
| -10            | 1.17            | 0.91           |
| -5             | 1.57            | 0.92           |
| -2             | 1.67            | 1.25           |
| 0              | 1.13            | 1.16           |
| +5             | 1.06            | 0.88           |

**Near-expiry test** (days  $-5$  to  $0$ ): High-HHI mean GK-vol = 1.39% vs 1.14%;  $t = 2.11$ ,  $p = 0.037$ . Contracts expiring in concentrated markets show 22% higher volatility in the final week.

Repeating the analysis with a herding regime split, divergent positioning (low herding) at expiry is associated with higher near-expiry GK-vol (1.37% vs 1.16%,  $p = 0.083$ ). The direction is consistent with the quintile results but only marginally significant, suggesting HHI is the more informative signal for expiry-window risk.

**Diff-in-diff control:** comparing near-expiry vol minus far-from-expiry vol across regimes is not significant for either HHI ( $p = 0.53$ ) or herding ( $p = 0.39$ ). The higher near-expiry vol in concentrated markets likely reflects a generally more volatile baseline rather than expiry-specific amplification.

## 2.3 Out-of-Sample Predictive Power

We train an expanding-window OLS (500-day minimum,  $\sim 1,250$  test days) and compare a model that includes the positioning variables against a baseline using only lagged realised vol:

The market-structure variables add genuine predictive power at the 20-day horizon—a +9.7 pp lift in out-of-sample  $R^2$  over a realised-vol-only baseline. At shorter horizons, the marginal contribution is negligible, consistent with the weekly cadence of the underlying positioning data.

**Table 3:** Out-of-sample  $R^2$  comparison

| Target       | Narrow model | Baseline (rvol only) | $\Delta$ |
|--------------|--------------|----------------------|----------|
| 1d fwd  ret  | -0.03        | 0.00                 | -2.5 pp  |
| 5d fwd rvol  | 0.04         | 0.04                 | +0.4 pp  |
| 20d fwd rvol | 0.14         | 0.05                 | +9.7 pp  |

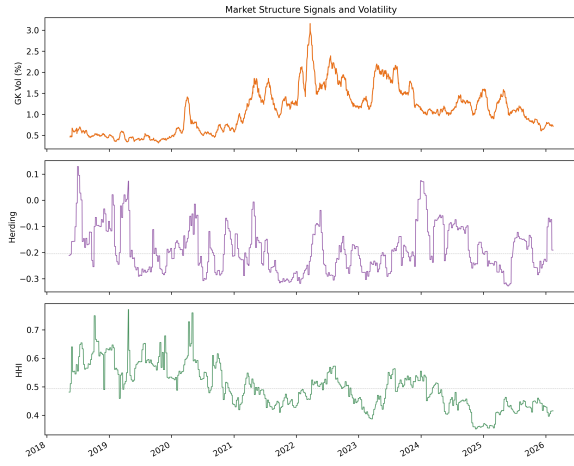
## 2.4 Robustness

The main results survive several stress tests. Bootstrapping the herding-range relationship (2,000 resamples) produces a negative coefficient in every draw (95% CI:  $[-0.027, -0.015]$ ). At weekly frequency—where each row is a genuinely new positioning observation rather than a forward-filled daily repeat—herding still leads volatility ( $p = 0.031$ ) and the out-of-sample improvement actually widens (+15.1 pp vs +9.7 pp daily). Adding seasonality and vol-of-vol controls with autocorrelation-robust standard errors leaves herding strongly significant ( $p < 0.001$ ); HHI holds for 20-day forward vol ( $p = 0.028$ ) but weakens for daily range, consistent with it being more of a structural than a high-frequency signal.

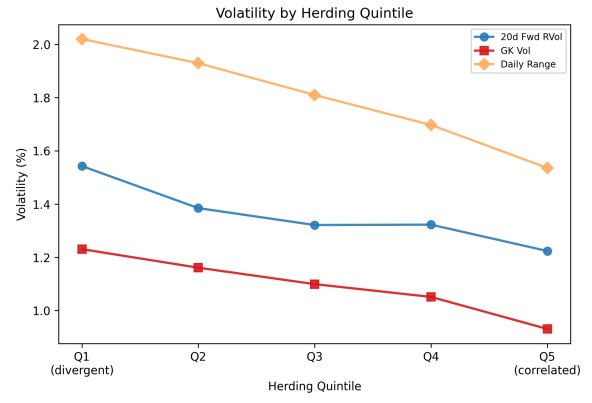
## 3 Practical Implications

1. **Market structure as a vol signal.** The HHI z-score and herding measure, both computable from weekly Euronext positioning reports, appear to contain forward-looking volatility information.
2. **Spread management.** Low-herding regimes (participant categories diverge) are associated with wider intraday ranges and higher realised vol over the following 20 days. This relationship is consistent across quintiles and survives robustness checks, though the sample is limited to Euronext rapeseed.
3. **Roll and expiry risk.** When HHIz13 is elevated ahead of contract expiry, the data shows  $\sim 22\%$  higher volatility in the final week ( $p = 0.037$ ). However, the diff-in-diff control is not significant, so this likely reflects generally higher baseline vol in concentrated periods rather than a purely expiry-specific effect.
4. **Vol regime monitoring.** Adding HHI, HHIz13, herding, and turnover ratio to a realised-vol baseline lifts the 20-day out-of-sample  $R^2$  by 9.7 pp. These variables are derived from publicly available weekly positioning reports and could be incorporated into existing vol models at minimal cost.
5. **The two signals complement each other.** Herding is the more robust general-purpose vol signal. HHI remains the more informative signal for expiry-window risk. Both are useful for volatility forecasting.

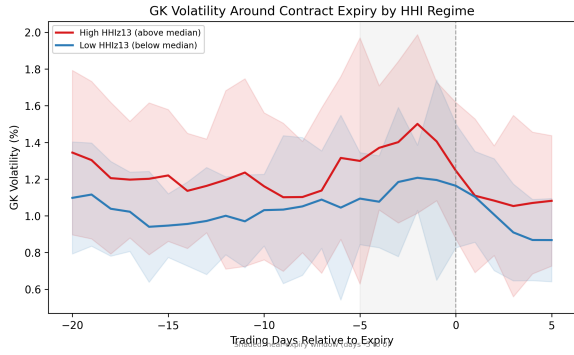
## A Figures



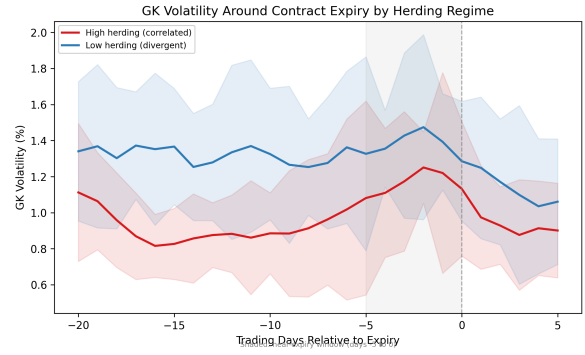
(a) Signals and GK vol (20d MA).



(b) Vol by herding quintile.



(c) Expiry event study – HHI regime.



(d) Expiry event study – herding regime.

**Figure 1:** Market structure and volatility. (a) GK vol, herding, and HHI time series. (b) Volatility by herding quintile (Q1 = divergent, Q5 = correlated). (c, d) GK vol around contract expiry split by regime; grey = near-expiry window.